

The Time Value of Money

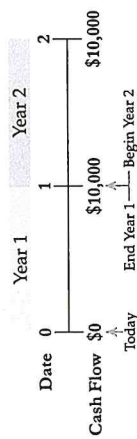
NOTATION

r	interest rate
C	cash flow
FV_n	future value on date n
PV_n	present value; annuity spreadsheet notation for the initial amount
C_n	cash flow at date n
N	date of the last cash flow in a stream of cash flows
NPV	net present value
P	initial principal or deposit, or equivalent present value
FV	future value; annuity spreadsheet notation for an extra final payment
g	growth rate
$NPER$	annuity spreadsheet notation for the number of periods or date of the last cash flow
$RATE$	annuity spreadsheet notation for interest rate
PMT	annuity spreadsheet notation for cash flow
IRR	internal rate of return
PV_n	present value on date n

4.1 The Timeline

We begin our look at valuing cash flows lasting several periods with some basic vocabulary and tools. We refer to a series of cash flows lasting several periods as a **stream of cash flows**. We can represent a stream of cash flows on a **timeline**, a linear representation of the timing of the expected cash flows. Timelines are an important first step in organizing and then solving a financial problem. We use them throughout this text.

To illustrate how to construct a timeline, assume that a friend owes you money. He has agreed to repay the loan by making two payments of \$10,000 at the end of each of the next two years. We represent this information on a timeline as follows:

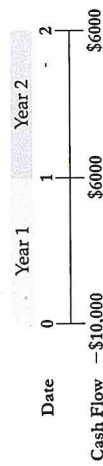


Date 0 represents the present. Date 1 is one year later and represents the end of the first year. The \$10,000 cash flow below date 1 is the payment you will receive at the end of the first year. Date 2 is two years from now; it represents the end of the second year. The \$10,000 cash flow below date 2 is the payment you will receive at the end of the second year.

To track cash flows on the timeline, interpret each point on the timeline as a specific date. The space between date 0 and date 1 then represents the time period between these dates—in this case, the first year of the loan. Date 0 is the beginning of the first year, and date 1 is the end of the first year. Similarly, date 1 is the beginning of the second year, and date 2 is the end of the second year. By denoting time in this way, date 1 signifies *both* the end of year 1 and the beginning of year 2, which makes sense since those dates are effectively the same point in time.¹

In this example, both cash flows are inflows. In many cases, however, a financial decision will involve both inflows and outflows. To differentiate between the two types of cash flows, we assign a different sign to each: Inflows are positive cash flows, whereas outflows are negative cash flows.

To illustrate, suppose you're still feeling generous and have agreed to lend your brother \$10,000 today. Your brother has agreed to repay this loan in two installments of \$6000 at the end of each of the next two years. The timeline is as follows:



Notice that the first cash flow at date 0 (today) is represented as -\$10,000 because it is an outflow. The subsequent cash flows of \$6000 are positive because they are inflows.

¹ That is, there is no real time difference between a cash flow paid at 11:59 p.m. on December 31 and one paid at 12:01 a.m. on January 1, although there may be some other differences such as taxation that we overlook for now.

AS DISCUSSED IN CHAPTER 3, TO EVALUATE A PROJECT, A FINANCIAL manager must compare its costs and benefits. In most cases, these costs and benefits are spread across time. For example, in 2016 Apple significantly increased R&D spending, topping \$10 billion per year for the first time. This growth in R&D expense has continued (spending nearly \$22 billion on R&D in 2021), likely reflecting the company's intention to develop new revolutionary products rather than just improving existing projects. Indeed, as famously secretive as Apple is, CEO Tim Cook openly touts Augmented Reality and has explicitly stated that Apple has "put a lot of energy" into the technology. Apple's R&D expenditures represent significant up-front costs. Executives hope these investments will translate into revenues that will occur many years or even decades in the future. How do financial managers compare costs and benefits that occur over many years?

In order to evaluate a long-term project such as an investment in R&D, we need tools that allow us to compare cash flows that occur at different points in time. We develop these tools in this chapter. The first tool is a visual method for representing a stream of cash flows: the timeline. After constructing a timeline, we establish three important rules for moving cash flows to different points in time. Using these rules, we show how to compute the present and future values of the costs and benefits of a general stream of cash flows. By converting all cash flows to a common point in time, these tools allow us to compare the costs and benefits of a long-term project, and thus assess its net present value, or NPV. The NPV expresses the net benefit of the project in terms of cash today.

While the general techniques developed in this chapter can be used to value any type of asset, certain types of assets have cash flows that follow a regular pattern. We develop shortcuts for valuing *annuities*, *perpetuities*, and other special cases of assets with cash flows that follow regular patterns.

So far, we have used timelines to show the cash flows that occur at the end of each year. Actually, timelines can represent cash flows that take place at the end of any time period. For example, if you pay rent each month, you could use a timeline like the one in our first example to represent two rental payments, but you would replace the “year” label with “month.”

Many of the timelines included in this chapter are very simple. Consequently, you may feel that it is not worth the time or trouble to construct them. As you progress to more difficult problems, however, you will find that timelines identify events in a transaction or investment that are easy to overlook. If you fail to recognize these cash flows, you will make flawed financial decisions. Therefore, we recommend that you approach every problem by drawing the timeline as we do in this chapter.

EXAMPLE 4.1

Constructing a Timeline

Problem

Suppose you must pay tuition of \$10,000 per year for the next two years. Your tuition payments must be made in equal installments at the start of each semester. What is the timeline of your tuition payments?

Solution

Assuming today is the start of the first semester, your first payment occurs at date 0 (today). The remaining payments occur at semester intervals. Using one semester as the period length, we can construct a timeline as follows:



CONCEPT CHECK

1. What are the key elements of a timeline?
2. How can you distinguish cash inflows from outflows on a timeline?

4.2 The Three Rules of Time Travel

Financial decisions often require comparing or combining cash flows that occur at different points in time. In this section, we introduce three important rules central to financial decision making that allow us to compare or combine values.

Rule 1: Comparing and Combining Values

Our first rule is that it is only possible to compare or combine values at the same point in time. This rule restates a conclusion introduced in Chapter 3: Only cash flows in the same units can be compared or combined. *A dollar today and a dollar in one year are not equivalent.* Having money now is more valuable than having money in the future; if you have the money today you can earn interest on it.

To compare or combine cash flows that occur at different points in time, you first need to convert the cash flows into the same units or *move* them to the same point in time. The next two rules show how to move the cash flows on the timeline.

Rule 2: Moving Cash Flows Forward in Time

Suppose we have \$1000 today, and we wish to determine the equivalent amount in one year's time. If the current market interest rate is 10%, we can use that rate as an exchange rate to move the cash flow forward in time. That is,

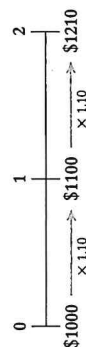
$$(\$1000 \text{ today}) \times (1.10 \$ \text{ in one year} / \$ \text{ today}) = \$1100 \text{ in one year}$$

In general, if the market interest rate for the year is r , then we multiply by the interest rate factor, $(1 + r)$, to move the cash flow from the beginning to the end of the year. This process of moving a value or cash flow forward in time is known as **compounding**. *Our second rule stipulates that to move a cash flow forward in time, you must compound it.*

We can apply this rule repeatedly. Suppose we want to know how much the \$1000 is worth in two years' time. If the interest rate for year 2 is also 10%, then we convert as we just did:

$$(\$1100 \text{ in one year}) \times (1.10 \$ \text{ in two years} / \$ \text{ in one year}) = \$1210 \text{ in two years}$$

Let's represent this calculation on a timeline as follows:



Given a 10% interest rate, all of the cash flows—\$1000 at date 0, \$1100 at date 1, and \$1210 at date 2—are equivalent. They have the same value but are expressed in different units (different points in time). An arrow that points to the right indicates that the value is being moved forward in time—that is, compounded.

The value of a cash flow that is moved forward in time is known as its future value. In the preceding example, \$1210 is the future value of \$1000 two years from today. Note that the value grows as we move the cash flow further in the future. The difference in value between money today and money in the future represents the **time value of money**, and it reflects the fact that by having money sooner, you can invest it and have more money later as a result. Note also that the equivalent value grows by \$100 the first year, but by \$110 the second year. In the second year we earn interest on our original \$1000, plus we earn interest on the \$100 interest we received in the first year. This effect of earning “interest on interest” is known as **compound interest**.

How does the future value change if we move the cash flow three years? Continuing with the same approach, we compound the cash flow a third time. Assuming the competitive market interest rate is fixed at 10%, we get

$$\$1000 \times (1.10) \times (1.10) \times (1.10) = \$1000 \times (1.10)^3 = \$1331$$

In general, to take a cash flow C forward n periods into the future, we must compound it by the n intervening interest rate factors. If the interest rate r is constant, then

Future Value of a Cash Flow

$$FV_n = C \times \underbrace{(1 + r) \times (1 + r) \times \cdots \times (1 + r)}_{n \text{ times}} = C \times (1 + r)^n \quad (4.1)$$

FIGURE 4.1

The Composition of Interest over Time

This graph shows the account balance and the composition of interest over time starting with an initial deposit of \$1,000, in an account paying 10% annual interest. The green (middle) bars show **simple interest**, the interest earned only on the initial deposit. The blue (top) bars show **compound interest**, the interest earned on prior interest payments. Note the increasing importance of compound interest over time.

An interactive version of this figure is available in the eTextbook and MyLab Finance.

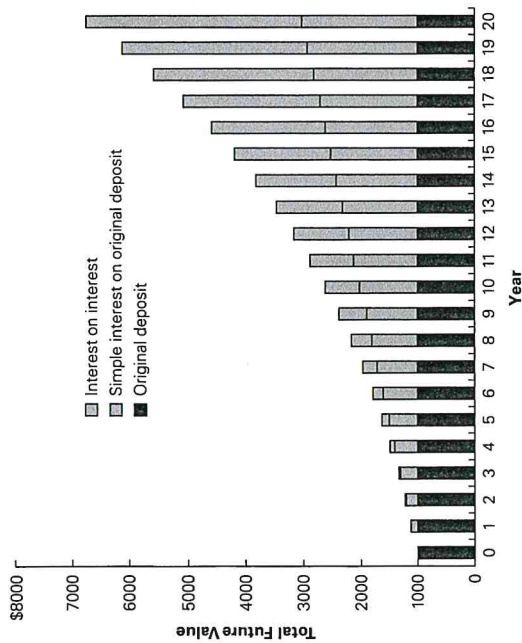


Figure 4.1 shows the importance of earning “interest on interest” in the growth of the account balance over time. The type of growth that results from compounding is called geometric or exponential growth. As Example 4.2 shows, over a long horizon, the effect of compounding can be quite dramatic.

EXAMPLE 4.2

The Power of Compounding

Problem

Suppose you invest \$1000 in an account paying 10% interest per year. How much will you have in the account in 7 years? in 20 years? in 75 years?

Solution

You can apply Eq. 4.1 to calculate the future value in each case:

$$7 \text{ years: } \$1000 \times (1.10)^7 = \$1948.72$$

$$20 \text{ years: } \$1000 \times (1.10)^{20} = \$6727.50$$

$$75 \text{ years: } \$1000 \times (1.10)^{75} = \$1,271,895.37$$

Note that at 10% interest, your money will nearly double in 7 years. After 20 years, it will increase almost seven-fold. And if you invest for 75 years, you will be a millionaire!

Rule 3: Moving Cash Flows Back in Time

The third rule describes how to move cash flows backward in time. Suppose you would like to compute the value today of \$1000 you anticipate receiving in one year. If the current

Rule of 72

Another way to think about the effect of compounding and discounting is to consider how long it will take your money to double given different interest rates. Suppose we want to know how many years it will take for \$1 to grow to a future value of \$2. We want the number of years, N , to solve

$$FV = \$1 \times (1 + r)^N = \$2$$

If you solve this formula for different interest rates, you will find the following approximation:

Years to double $\approx 72 \div (\text{interest rate in percent})$

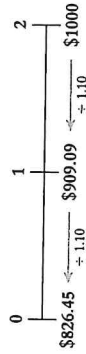
This simple “Rule of 72” is fairly accurate (i.e., within one year of the exact doubling time) for interest rates higher than 2%. For example, if the interest rate is 9%, the doubling time should be about $72 \div 9 = 8$ years. Indeed, $1.09^8 = 1.991$. So, given a 9% interest rate, your money will approximately double every eight years.²

market interest rate is 10%, you can compute this value by converting units as we did in Chapter 3:

$$(\$1000 \text{ in one year}) \div (1.10 \$ \text{ in one year}/\$ \text{ today}) = \$909.09 \text{ today}$$

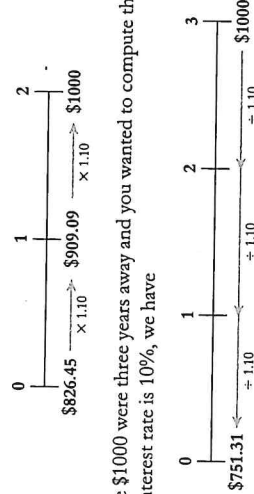
That is, to move the cash flow backward in time, we divide it by the interest rate factor, $(1 + r)$, where r is the interest rate. This process of moving a value or cash flow backward in time—finding the equivalent value today of a future cash flow—is known as **discounting**. Our third rule stipulates that to move a cash flow back in time, we must discount it.

To illustrate, suppose that you anticipate receiving the \$1000 two years from today rather than in one year. If the interest rate for both years is 10%, we can prepare the following timeline:



When the interest rate is 10%, all of the cash flows—\$826.45 at date 0, \$909.09 at date 1, and \$1000 at date 2—are equivalent. They represent the same value in different units (different points in time). The arrow points to the left to indicate that the value is being moved backward in time or discounted. Note that the value decreases as we move the cash flow further back.

The value of a future cash flow at an earlier point on the timeline is its present value at the earlier point in time. That is, \$826.45 is the present value at date 0 of \$1000 in two years. Recall from Chapter 3 that the present value is the “do-it-yourself” price to produce a future cash flow. Thus, if we invested \$826.45 today for two years at 10% interest, we would have a future value of \$1000, using the second rule of time travel:



Suppose the \$1000 were three years away and you wanted to compute the present value. Again, if the interest rate is 10%, we have

² See the appendix to this chapter for an explanation of how to calculate the exact doubling time.

That is, the present value today of a cash flow of \$1000 in three years is given by

$$\$1000 \div (1.10) + (1.10) \div (1.10) = \$1000 \div (1.10)^3 = \$751.31$$

In general, to move a cash flow C backward n periods, we must discount it by the n intervening interest rate factors. If the interest rate r is constant, then

Present Value of a Cash Flow

$$PV = C \div (1+r)^n = \frac{C}{(1+r)^n} \quad (4.2)$$

EXAMPLE 4.3

Present Value of a Single Future Cash Flow

Problem

You are considering investing in a savings bond that will pay \$15,000 in 10 years. If the competitive market interest rate is fixed at 6% per year, what is the bond worth today?

Solution

The cash flows for this bond are represented by the following timeline:



Thus, the bond is worth \$15,000 in 10 years. To determine the value today, we compute the present value:

$$PV = \frac{15,000}{1.06^{10}} = \$8375.92 \text{ today}$$

The bond is worth much less today than its final payoff because of the time value of money.

Applying the Rules of Time Travel

The rules of time travel allow us to compare and combine cash flows that occur at different points in time. Suppose we plan to save \$1000 today, and \$1000 at the end of each of the next two years. If we earn a fixed 10% interest rate on our savings, how much will we have three years from today?

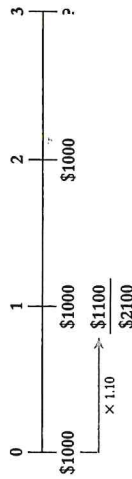
Again, we start with a timeline:



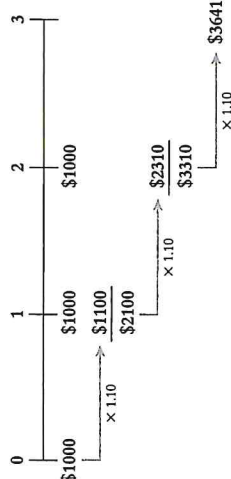
The timeline shows the three deposits we plan to make. We need to compute their value at the end of three years.

We can use the rules of time travel in a number of ways to solve this problem. First, we can take the deposit at date 0 and move it forward to date 1. Because it is then in the same

time period as the date 1 deposit, we can combine the two amounts to find out the total in the bank on date 1:

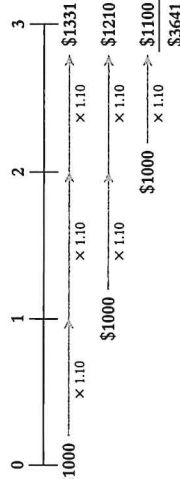


Using the first two rules of time travel, we find that our total savings on date 1 will be \$2100. Continuing in this fashion, we can solve the problem as follows:



The total amount we will have in the bank at the end of three years is \$3641. This amount is the future value of our \$1000 savings deposits.

Another approach to the problem is to compute the future value in year 3 of each cash flow separately. Once all three amounts are in year 3 dollars, we can then combine them.



Both calculations give the same future value. As long as we follow the rules, we get the same result. The order in which we apply the rules does not matter. The calculation we choose depends on which is more convenient for the problem at hand. Table 4.1 summarizes the three rules of time travel and their associated formulas.

TABLE 4.1

The Three Rules of Time Travel

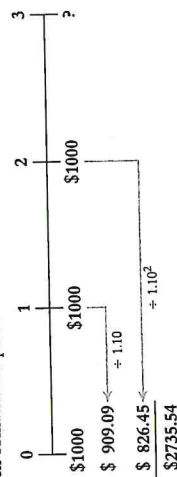
Rule 1	Only values at the same point in time can be compared or combined.	Future Value of a Cash Flow $FV_n = C \times (1+r)^n$
Rule 2	To move a cash flow forward in time, you must compound it.	Present Value of a Cash Flow $PV = C \div (1+r)^n$
Rule 3	To move a cash flow backward in time, you must discount it.	

EXAMPLE 4.4 Computing the Future Value**Problem**

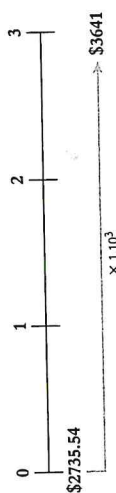
Let's revisit the savings plan we considered earlier: We plan to save \$1000 today and at the end of each of the next two years. At a fixed 10% interest rate, how much will we have in the bank three years from today?

Solution

Let's solve this problem in a different way than we did earlier. First, compute the present value of the cash flows. There are several ways to perform this calculation. Here we treat each cash flow separately and then combine the present values.



Saving \$2735.54 today is equivalent to saving \$1000 per year for three years. Now let's compute its future value in year 3:



This answer of \$3641 is precisely the same result we found earlier. As long as we apply the three rules of time travel, we will always get the correct answer.

1. Can you compare or combine cash flows at different times?
2. What is compound interest?
3. How do you move a cash flow backward and forward in time?

4.3 Valuing a Stream of Cash Flows

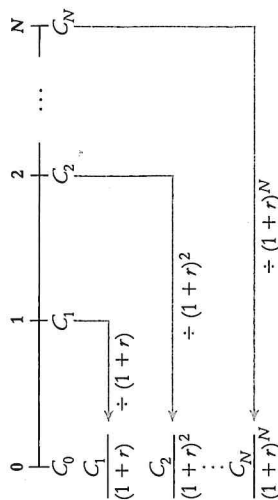
Most investment opportunities have multiple cash flows that occur at different points in time. In Section 4.2, we applied the rules of time travel to value such cash flows. Now, we formalize this approach by deriving a general formula for valuing a stream of cash flows.

Consider a stream of cash flows: C_0 at date 0, C_1 at date 1, and so on, up to C_N at date N . We represent this cash flow stream on a timeline as follows:



Using the time travel techniques, we compute the present value of this cash flow stream in two steps. First, we compute the present value of each individual cash flow. Then, once the cash flows are in common units of dollars today, we can combine them.

For a given interest rate r , we represent this process on the timeline as follows:



This timeline provides the general formula for the present value of a cash flow stream:

$$PV = C_0 + \frac{C_1}{(1+r)} + \frac{C_2}{(1+r)^2} + \dots + \frac{C_N}{(1+r)^N} \quad (4.3)$$

We can also write this formula as a summation:

Present Value of a Cash Flow Stream

$$PV = \sum_{n=0}^N PV(C_n) = \sum_{n=0}^N \frac{C_n}{(1+r)^n} \quad (4.4)$$

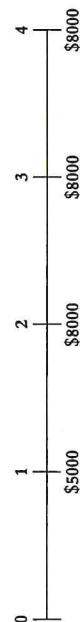
The summation sign, $\sum$, means "sum the individual elements for each date n from 0 to N ." Note that $(1+r)^0 = 1$, so this shorthand matches precisely Eq. 4.3. That is, the present value of the cash flow stream is the sum of the present values of each cash flow. Recall from Chapter 3 how we defined the present value as the dollar amount you would need to invest today to produce the single cash flow in the future. The same idea holds in this context. The present value is the amount you need to invest today to generate the cash flow stream $C_0, C_1, \dots, C_N$. That is, receiving those cash flows is equivalent to having their present value in the bank today.

EXAMPLE 4.5 Present Value of a Stream of Cash Flows**Problem**

You have just graduated and need money to buy a new car. Your rich Uncle Henry will lend you the money so long as you agree to pay him back within four years, and you offer to pay him the rate of interest that he would otherwise get by putting his money in a savings account. Based on your earnings and living expenses, you think you will be able to pay him \$5000 in one year, and then \$8000 each year for the next three years. If Uncle Henry would otherwise earn 6% per year on his savings, how much can you borrow from him?

Solution

The cash flows you can promise Uncle Henry are as follows:



How much money should Uncle Henry be willing to give you today in return for your promise of these payments? He should be willing to give you an amount that is equivalent to these

payments in present value terms. This is the amount of money that it would take him to produce these same cash flows, which we calculate as follows:

$$\begin{aligned} PV &= \frac{5000}{1.06} + \frac{8000}{1.06^2} + \frac{8000}{1.06^3} + \frac{8000}{1.06^4} \\ &= 4716.98 + 7119.97 + 6716.95 + 6336.75 \\ &= 24,890.65 \end{aligned}$$

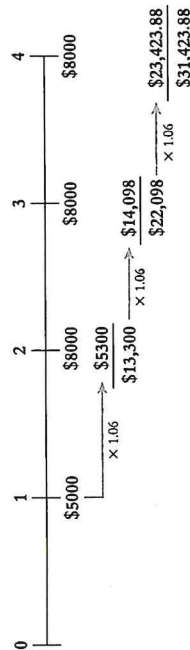
Thus, Uncle Henry should be willing to lend you \$24,890.65 in exchange for your promised payments. This amount is less than the total you will pay him (\$5000 + \$8000 + \$8000 + \$8000 = \$29,000) due to the time value of money.

Let's verify our answer. If your uncle kept his \$24,890.65 in the bank today earning 6% interest, in four years he would have

$$FV = \$24,890.65 \times (1.06)^4 = \$31,423.87 \text{ in four years}$$

Now suppose that Uncle Henry gives you the money, and then deposits your payments to him in the bank each year. How much will he have four years from now?

We need to compute the future value of the annual deposits. One way to do so is to compute the bank balance each year:



We get the same answer both ways (within a penny, which is because of rounding).

The last section of Example 4.5 illustrates a general point. If you want to compute the future value of a stream of cash flows, you can do it directly (the second approach used in Example 4.5), or you can first compute the present value and then move it to the future (the first approach). Because we obey the laws of time travel in both cases, we get the same result. This principle can be applied more generally to write the following formula for the future value in year n in terms of the present value of a set of cash flows:

Future Value of a Cash Flow Stream with a Present Value of PV

$$FV_n = PV \times (1 + r)^n \quad (4.5)$$

1. How do you calculate the present value of a cash flow stream?
2. How do you calculate the future value of a cash flow stream?

4.4 Calculating the Net Present Value

Now that we have established the rules of time travel and determined how to compute present and future values, we are ready to address our central goal: comparing the costs and benefits of a project to evaluate a long-term investment decision. From our first rule of time travel, to compare cash flows we must value them at a common point in time. A

convenient choice is to use present values. In particular, we define the **net present value (NPV)** of an investment decision as follows:

$$NPV = PV(\text{benefits}) - PV(\text{costs}) \quad (4.6)$$

In this context, the benefits are the cash inflows and the costs are the cash outflows. We can represent any investment decision on a timeline as a cash flow stream where the cash outflows (investments) are negative cash flows and the inflows are positive cash flows. Thus, the NPV of an investment opportunity is also the *present value* of the stream of cash flows of the opportunity.

$$NPV = PV(\text{benefits}) - PV(\text{costs}) = PV(\text{benefits} - \text{costs})$$

EXAMPLE 4.6 Net Present Value of an Investment Opportunity

Problem

You have been offered the following investment opportunity: If you invest \$1000 today, you will receive \$500 at the end of each of the next three years. If you could otherwise earn 10% per year on your money, should you undertake the investment opportunity?

Solution

As always, we start with a timeline. We denote the up-front investment as a negative cash flow (because it is money we need to spend) and the money we receive as a positive cash flow.



To decide whether we should accept this opportunity, we compute the NPV by computing the present value of the stream:

$$NPV = -1000 + \frac{500}{1.10} + \frac{500}{1.10^2} + \frac{500}{1.10^3} = \$243.43$$

Because the NPV is positive, the benefits exceed the costs and we should make the investment. Indeed, the NPV tells us that taking this opportunity is like getting an extra \$243.43 that you can spend today. To illustrate, suppose you borrow \$1000 to invest in the opportunity and an extra \$243.43 to spend today. How much would you owe on the \$1243.43 loan in three years? At 10% interest, the amount you would owe would be

$$FV = (\$1000 + \$243.43) \times (1.10)^3 = \$1655 \text{ in three years}$$

At the same time, the investment opportunity generates cash flows. If you put these cash flows into a bank account, how much will you have saved three years from now? The future value of the savings is

$$FV = (\$500 \times 1.10^2) + (\$500 \times 1.10) + \$500 = \$1655 \text{ in three years}$$

As you see, you can use your bank savings to repay the loan. Taking the opportunity therefore allows you to spend \$243.43 today at no extra cost.

In principle, we have explained how to answer the question we posed at the beginning of the chapter: How should financial managers evaluate the cash flows from undertaking a multiyear project like Apple's investment in R&D? We have shown how to compute the NPV of an investment opportunity that lasts more than one period. In practice, when the number of cash flows exceeds four or five (as it most likely will), the calculations can become tedious. Fortunately, a number of special cases do not require us to treat each cash flow separately. We derive these shortcuts in Section 4.5.

USING EXCEL

Calculating Present Values in Excel

Calculating NPV

While present and future value calculations can be done with a calculator, it is often convenient to evaluate them using a spreadsheet program. For example, the following spreadsheet calculates the NPV in Example 4.6:

A	B	C	D	E
1	Discount Rate	10.0%		
2	Period	0	1	2
3	Cash Flow C_t	(1,000.0)	500.0	500.0
4	Discount Factor	1.000	0.909	0.826
5	$PV(C_t)$	(1,000.0)	454.5	413.2
6	NPV	243.43		

Rows 1–3 provide the key data of the problem, the discount rate, and the cash flow timeline (note we use a blue font to indicate input data, and black for cells that are fixed or calculated). Row 4 then calculates the discount factor, $1/(1+r)^t$, the present value of a dollar received in year t . We multiply each cash flow by the discount factor to convert it to a present value, shown in row 5. Finally, row 6 shows the sum of the present values of all the cash flows, which is the NPV. The formulas in rows 4–6 are shown below:

A	B	C	D	E
4	Discount Factor	$=1/(1+\$B\$1)^{B2}$	$=1/(1+\$B\$1)^{D2}$	$=1/(1+\$B\$1)^{E2}$
5	$PV(C_t)$	$=B3*B4$	$=C3*C4$	$=E3*E4$
6	NPV	$=SUM(B5:E5)$		

Alternatively, we could have computed the entire NPV in one step, using a single (long) formula. We recommend as a best practice that you avoid that temptation and calculate the NPV step by step. Doing so facilitates error checking and makes clear the contribution of each cash flow to the overall NPV.

Excel's NPV Function

Excel also has a built-in NPV function. This function has the format $NPV(\text{rate}, \text{value1}, \text{value2}, \dots)$, where “rate” is the interest rate per period used to discount the cash flows, and “value1”, “value2”, and so on are the cash flows (or ranges of cash flows). Unfortunately, however, the NPV function computes the present value of the cash flows *assuming the first cash flow occurs at date 1*. Therefore, if a project's first cash flow occurs at date 0, we must add it separately. For example, we could calculate NPV in the spreadsheet above with the formula

$$= B3 + NPV(B1, C3:E3)$$

Another pitfall with the NPV function is that cash flows that are left blank are treated differently from cash flows that are equal to zero. If the cash flow is left blank, *both the cash flow and the period are ignored*. For example, compare the results below if we delete the period 2 cash flow:

A	B	C	D	E
1	Discount Rate	10.0%		
2	Period	0	1	3
3	Cash Flow C_t	(1,000.0)	500.0	500.0
4	Discount Factor	1.000	0.909	0.826
5	$PV(C_t)$	(1,000.0)	454.5	375.7
6	NPV	$=SUM(B5:E5)$		
7	NPV function	$=B3+NPV(B1,C3:E3)$		

Our original method provides the correct solution in row 6, whereas the NPV function used in row 7 treats the cash flow in period 3 as though it occurred at period 2, which is clearly not what is intended and is incorrect.

CONCEPT CHECK

1. How do you calculate the net present value of a cash flow stream?
2. What benefit does a firm receive when it accepts a project with a positive NPV?

4.5 Perpetuities and Annuities

The formulas we have developed so far allow us to compute the present or future value of any cash flow stream. In this section, we consider two special types of cash flow streams, *perpetuities* and *annuities*, and we learn shortcuts for valuing them. These shortcuts are possible because the cash flows follow a regular pattern.

Perpetuities

A *perpetuity* is a stream of equal cash flows that occur at regular intervals and last forever. One example is the British government bond called a *consol* (or perpetual bond). Consol bonds promise the owner a fixed cash flow every year, forever.

Here is the timeline for a perpetuity:



Note from the timeline that the first cash flow does not occur immediately; it *arrives at the end of the first period*. This timing is sometimes referred to as *payment in arrears* and is a standard convention that we adopt throughout this text.

Using the formula for the present value, the present value of a perpetuity with payment C and interest rate r is given by

$$PV = \frac{C}{(1+r)} + \frac{C}{(1+r)^2} + \frac{C}{(1+r)^3} + \dots = \sum_{n=1}^{\infty} \frac{C}{(1+r)^n}$$

Notice that $C_n = C$ in the present value formula because the cash flow for a perpetuity is constant. Also, because the first cash flow is in one period, $C_0 = 0$.

To find the value of a perpetuity one cash flow at a time would take forever—literally! You might wonder how, even with a shortcut, the sum of an infinite number of positive terms

Historical Examples of Perpetuities

Companies sometimes issue bonds that they call *perpetuities*, but in fact are not really perpetuities. For example, in mid-2010, Europe's largest bank, HSBC, sold \$3.4 billion of “perpetual” bonds that promise investors a fixed amount each year with no maturity date. But while the bonds have no fixed maturity, they are not exactly true perpetuities as HSBC has the right to pay off the bonds after 5½ years. Thus, the bond's payments might not last forever.

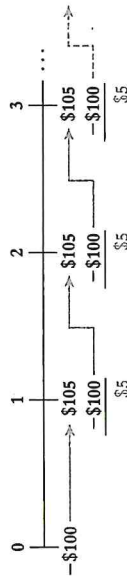
Perpetual bonds were some of the first bonds ever issued. The oldest perpetuities that are still making interest payments were issued in 1624 by the *Hoogheemraadschap Leidsche Bovenwaters*, a seventeenth-century Dutch water board responsible for upkeep of the local dikes. To verify that these bonds continue to pay interest, two finance professors at Yale University, William Goetzmann and Geert

Rouwenhorst, purchased one of these bonds in July 2003, and collected 26 years of back interest. On its issue date in 1648, this bond originally paid interest in Carolus guilders. Over the next 355 years, the currency of payment changed to Flemish pounds, Dutch guilders, and most recently euros. Currently, the bond pays interest of €11.34 annually.

Although the Dutch bonds are the oldest perpetuities still in existence, the first perpetuities date from much earlier times. For example, *cenus agreements* and *rentes*, which were forms of perpetuities and annuities, were issued in the twelfth century in Italy, France, and Spain. They were initially designed to circumvent the usury laws of the Catholic Church: Because they did not require the repayment of principal, in the eyes of the church they were not considered loans.

could be finite. The answer is that the cash flows in the future are discounted for an ever-increasing number of periods, so their contribution to the sum eventually becomes negligible.³

To derive the shortcut, we calculate the value of a perpetuity by creating our own perpetuity. We can then calculate the present value of the perpetuity because, by the Law of One Price, the value of the perpetuity must be the same as the cost we would incur to create it ourselves. To illustrate, suppose you could invest \$100 in a bank account paying 5% interest per year forever. At the end of one year, you will have \$105 in the bank—your original \$100 plus 5% interest. Suppose you withdraw the \$5 interest and reinvest the \$100 for a second year. Again you will have \$105 after one year, and you can withdraw \$5 and reinvest \$100 for another year. By doing this year after year, you can withdraw \$5 every year in perpetuity:



By investing \$100 in the bank today, you can, in effect, create a perpetuity paying \$5 per year. The Law of One Price tells us that the same good must have the same price in every market. Because the bank will “sell” us (allow us to create) the perpetuity for \$100, the present value of the \$5 per year in perpetuity is this “do-it-yourself” cost of \$100.

Now let's generalize this argument. Suppose we invest an amount P in the bank. Every year we can withdraw the interest we have earned, $C = r \times P$, leaving the principal, P , in the bank. The present value of receiving C in perpetuity is therefore the up-front cost $P = C/r$. Therefore,

Present Value of a Perpetuity

$$PV(C \text{ in perpetuity}) = \frac{C}{r} \quad (4.7)$$

In other words, by depositing the amount C/r today, we can withdraw interest of $(C/r) \times r = C$ each period in perpetuity. Thus, the present value of the perpetuity is C/r .

Note the logic of our argument. To determine the present value of a cash flow stream, we computed the “do-it-yourself” cost of creating those same cash flows at the bank. This is an extremely useful and powerful approach—and is much simpler and faster than summing those infinite terms!⁴

EXAMPLE 4.7

Endowing a Perpetuity

Problem

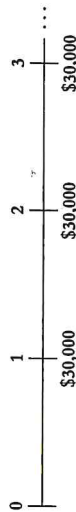
You want to endow an annual MBA graduation party at your alma mater. You want the event to be a memorable one, so you budget \$30,000 per year forever for the party. If the university earns 8% per year on its investments, and if the first party is in one year's time, how much will you need to donate to endow the party?

³ In mathematical terms, this is a geometric series, so it converges as long as $r > 0$.

⁴ Another mathematical derivation of this result exists (see online appendix at www.bekdemarzo.com), but it is less intuitive. This case is a good example of how the Law of One Price can be used to derive useful results.

Solution

The timeline of the cash flows you want to provide is



This is a standard perpetuity of \$30,000 per year. The funding you would need to give the university in perpetuity is the present value of this cash flow stream. From the formula,

$$PV = C/r = \$30,000/0.08 = \$375,000 \text{ today}$$

If you donate \$375,000 today, and if the university invests it at 8% per year forever, then the MBAs will have \$30,000 every year for their graduation party.

Annuities

An annuity is a stream of N equal cash flows paid at regular intervals. The difference between an annuity and a perpetuity is that an annuity ends after some fixed number of payments. Most car loans, mortgages, and some bonds are annuities. We represent the cash flows of an annuity on a timeline as follows.

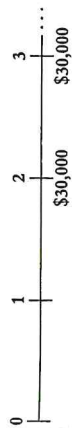


COMMON MISTAKE Discounting One Too Many Times

The perpetuity formula assumes that the first payment occurs at the end of the first period (at date 1). Sometimes perpetuities have cash flows that start later in the future. In this case, we can adapt the perpetuity formula to compute the present value, but we need to do so carefully to avoid a common mistake.

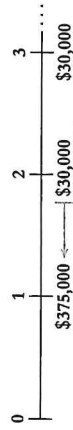
To illustrate, consider the MBA graduation party described in Example 4.7. Rather than starting immediately, suppose that the first party will be held two years from today (for the current entering class). How would this delay change the amount of the donation required?

Now the timeline looks like this:



We need to determine the present value of these cash flows, as it tells us the amount of money in the bank needed today to finance the future parties. We cannot apply the perpetuity formula directly, however, because these cash flows are not *exactly* a perpetuity as we defined it. Specifically, the cash flow in the first period is “missing.” But consider the

situation on date 1—at that point, the first party is one period away and then the cash flows are periodic. From the perspective of date 1, this *is* a perpetuity, and we can apply the formula. From the preceding calculation, we know we need \$375,000 on date 1 to have enough to start the parties on date 2. We rewrite the timeline as follows:



Our goal can now be restated more simply: How much do we need to invest today to have \$375,000 in one year? This is a simple present value calculation:

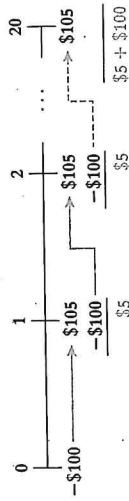
$$PV = \$375,000/1.08 = \$347,222 \text{ today}$$

A common mistake is to discount the \$375,000 twice because the first party is in two periods. Remember—the *present value formula for the perpetuity already discounts the cash flow to one period prior to the first cash flow*. Keep in mind that this common mistake may be made with perpetuities, annuities, and all of the other special cases discussed in this section. All of these formulas dis-

Note that just as with the perpetuity, we adopt the convention that the first payment takes place at date 1, one period from today. The present value of an N -period annuity with payment C and interest rate r is

$$PV = \frac{C}{(1+r)} + \frac{C}{(1+r)^2} + \frac{C}{(1+r)^3} + \cdots + \frac{C}{(1+r)^N} = \sum_{n=1}^N \frac{C}{(1+r)^n}$$

Present Value of an Annuity. To find a simpler formula, we use the same approach we followed with the perpetuity: find a way to create an annuity. To illustrate, suppose you invest \$100 in a bank account paying 5% interest. At the end of one year, you will have \$105 in the bank—your original \$100 plus \$5 in interest. Using the same strategy as for a perpetuity, suppose you withdraw the \$5 interest and reinvest the \$100 for a second year. Once again you will have \$105 after one year, and you can repeat the process, withdrawing \$5 and reinvesting \$100, every year. For a perpetuity, you left the principal in forever. Alternatively, you might decide after 20 years to close the account and withdraw the principal. In that case, your cash flows will look like this:



With your initial \$100 investment, you have created a 20-year annuity of \$5 per year, plus you will receive an extra \$100 at the end of 20 years. By the Law of One Price, because it took an initial investment of \$100 to create the cash flows on the timeline, the present value of these cash flows is \$100, or

$$\$100 = PV(\text{20-year annuity of \$5 per year}) + PV(\$100 \text{ in 20 years})$$

Rearranging terms gives

$$\begin{aligned} PV(\text{20-year annuity of \$5 per year}) &= \$100 - PV(\$100 \text{ in 20 years}) \\ &= 100 - \frac{100}{(1.05)^{20}} = \$62.31 \end{aligned}$$

So the present value of \$5 for 20 years is \$62.31. Intuitively, the value of the annuity is the initial investment in the bank account minus the present value of the principal that will be left in the account after 20 years.

We can use the same idea to derive the general formula. First, we invest P in the bank, and withdraw only the interest $C = r \times P$ each period. After N periods, we close the account. Thus, for an initial investment of P , we will receive an N -period annuity of C per period, plus we will get back our original P at the end. P is the total present value of the two sets of cash flows, or

$$P = PV(\text{annuity of } C \text{ for } N \text{ periods}) + PV(P \text{ in period } N)$$

By rearranging terms, we compute the present value of the annuity:

$$\begin{aligned} PV(\text{annuity of } C \text{ for } N \text{ periods}) &= P - PV(P \text{ in period } N) \\ &= P - \frac{P}{(1+r)^N} = P \left(1 - \frac{1}{(1+r)^N} \right) \end{aligned} \quad (4.8)$$

Recall that the periodic payment C is the interest earned every period; that is, $C = r \times P$ or, equivalently, solving for P provides the up-front cost in terms of C ,

$$P = C/r$$

Making this substitution for P in Eq. 4.8, provides the formula for the present value of an annuity of C for N periods.

Present Value of an Annuity⁵

$$PV(\text{annuity of } C \text{ for } N \text{ periods with interest rate } r) = C \times \frac{1}{r} \left(1 - \frac{1}{(1+r)^N} \right) \quad (4.9)$$

EXAMPLE 4.8

Present Value of an Annuity Due

Problem

You are the lucky winner of the \$30 million state lottery. You can take your prize money either as (a) 30 payments of \$1 million per year (starting today), or (b) \$15 million paid today. If the interest rate is 8%, which option should you take?

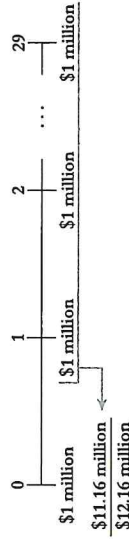
Solution

Option (a) provides \$30 million of prize money but paid annually. In this case, the cash flows are an annuity in which the first payment begins immediately, sometimes called an **annuity due**.⁶

Because the first payment is paid today, the last payment will occur in 29 years (for a total of 30 payments). We can compute the present value of the final 29 payments as a standard annuity of \$1 million per year using the annuity formula:

$$\begin{aligned} PV(\text{29 yr annuity of \$1 million/yr}) &= \$1 \text{ million} \times \frac{1}{.08} \left(1 - \frac{1}{1.08^{29}} \right) \\ &= \$11.16 \text{ million today} \end{aligned}$$

Adding the \$1 million we receive up front, this option has a present value of \$12.16 million:



Therefore, the present value of option (a) is only \$12.16 million, and so it is more valuable to take option (b) and receive \$15 million up front—even though we receive only half the total cash amount. The difference, of course, is due to the time value of money. To see that (b) really is better, if you have the \$15 million today, you can use \$1 million immediately and invest the remaining \$14 million at an 8% interest rate. This strategy will give you \$14 million $\times$ 8% = \$1.12 million per year in perpetuity! Alternatively, you can spend \$15 million – \$11.16 million = \$3.84 million today, and invest the remaining \$11.16 million, which will still allow you to withdraw \$1 million each year for the next 29 years before your account is depleted.

⁵ An early derivation of this formula is attributed to the astronomer Edmond Halley ("Of Compound Interest," published after Halley's death by Henry Sherwin, Sherwin's Mathematical Tables, London: W. and J. Mount, T. Page and Son, 1761).

⁶ Throughout the text, we will always use the term "annuity" on its own to mean one that is paid in arrears, starting at the end of the first period.

Formula for an Annuity Due

Although it is straightforward to calculate the value of an annuity due as we did in Example 4.8, it is not uncommon for practitioners to use the following equivalent formula:

$$PV(\text{annuity due of } C \text{ for } N \text{ periods}) = C \times \frac{1}{r} \left(1 - \frac{1}{(1+r)^N} \right) (1+r)$$

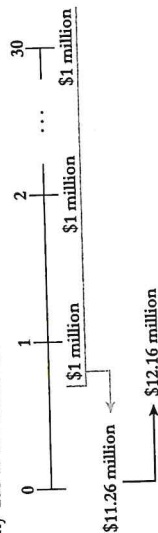
To understand where this formula comes from, note that one can think of an annuity due as the future value of a

regular annuity in one year. To illustrate, compute the present value of the regular annuity in Example 4.8:

$$PV(30 \text{ yr annuity of } \$1 \text{ million/yr}) =$$

$$\$1 \text{ million} \times \frac{1}{.08} \left(1 - \frac{1}{1.08^{30}} \right) = \$11.26 \text{ million}$$

Then compute the future value in a year: $\$11.26 \text{ million} \times 1.08 = \12.16 million . On a timeline it looks like this:



Future Value of an Annuity. Now that we have derived a simple formula for the present value of an annuity, it is easy to find a simple formula for the future value. If we want to know the value N years in the future, we move the present value N periods forward on the timeline; that is, we compound the present value for N periods at interest rate r :

Future Value of an Annuity

$$\begin{aligned} FV(\text{annuity}) &= PV \times (1+r)^N \\ &= \frac{C}{r} \left(1 - \frac{1}{(1+r)^N} \right) \times (1+r)^N \\ &= C \times \frac{1}{r} \left((1+r)^N - 1 \right) \end{aligned} \quad (4.10)$$

This formula is useful if we want to know how a savings account will grow over time. Let's apply this result to evaluate a retirement savings plan.

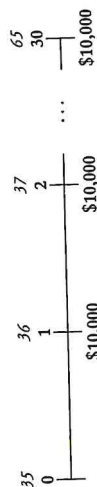
EXAMPLE 4.9 Retirement Savings Plan Annuity

Problem

Ellen is 35 years old, and she has decided it is time to plan seriously for her retirement. At the end of each year until she is 65, she will save \$10,000 in a retirement account. If the account earns 10% per year, how much will Ellen have saved at age 65?

Solution

As always, we begin with a timeline. In this case, it is helpful to keep track of both the dates and Ellen's age:



Ellen's savings plan looks like an annuity of \$10,000 per year for 30 years. (*Hint:* It is easy to become confused when you just look at age, rather than at both dates and age. A common error is to think there are only $65 - 36 = 29$ payments. Writing down both dates and age avoids this problem.)

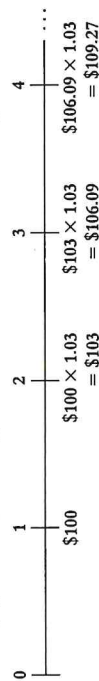
To determine the amount Ellen will have in the bank at age 65, we compute the future value of this annuity:

$$\begin{aligned} FV &= \$10,000 \times \frac{1}{0.10} (1.10^{30} - 1) \\ &= \$10,000 \times 164.49 \\ &= \$1.645 \text{ million at age 65} \end{aligned}$$

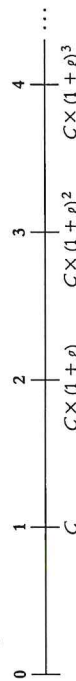
Growing Cash Flows

So far, we have considered only cash flow streams that have the same cash flow every period. If, instead, the cash flows are expected to grow at a constant rate in each period, we can also derive a simple formula for the present value of the future stream.

Growing Perpetuity. A growing perpetuity is a stream of cash flows that occur at regular intervals and grow at a constant rate forever. For example, a growing perpetuity with a first payment of \$100 that grows at a rate of 3% has the following timeline:



In general, a growing perpetuity with a first payment C and a growth rate g will have the following series of cash flows:



As with perpetuities with equal cash flows, we adopt the convention that the first payment occurs at date 1. Note a second important convention: *The first payment does not include growth.* That is, the first payment is C , even though it is one period away. Similarly, the cash flow in period n undergoes only $n - 1$ periods of growth. Substituting the cash flows from the preceding timeline into the general formula for the present value of a cash flow stream gives

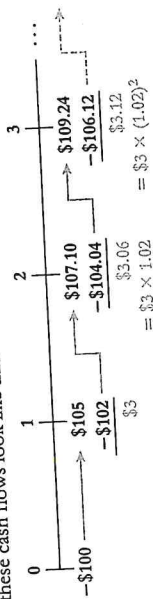
$$PV = \frac{C}{(1+r)} + \frac{C(1+g)}{(1+r)^2} + \frac{C(1+g)^2}{(1+r)^3} + \dots = \sum_{n=1}^{\infty} \frac{C(1+g)^{n-1}}{(1+r)^n}$$

Suppose $g \geq r$. Then the cash flows grow even faster than they are discounted; each term in the sum gets larger, rather than smaller. In this case, the sum is infinite! What does an infinite present value mean? Remember that the present value is the "do-it-yourself" cost of creating the cash flows. An infinite present value means that no matter how much money you start with, it is *impossible* to sustain a growth rate of g forever and reproduce those cash flows on your own. Growing perpetuities of this sort cannot exist in practice because no one would be willing to offer one at any finite price. A promise to pay an amount that forever grew faster than the interest rate is also unlikely to be kept (or believed by any savvy buyer).

The only viable growing perpetuities are those where the perpetual growth rate is less than the interest rate, so that each successive term in the sum is less than the previous term and the overall sum is finite. Consequently, we assume that $g < r$ for a growing perpetuity.

To derive the formula for the present value of a growing perpetuity, we follow the same logic used for a regular perpetuity: Compute the amount you would need to deposit today to create the perpetuity yourself. In the case of a regular perpetuity, we created a constant payment forever by withdrawing the interest earned each year and reinvesting the principal. To increase the amount we can withdraw each year, the principal that we reinvest each year must grow. Therefore, we withdraw less than the full amount of interest earned each period, using the remaining interest to increase our principal.

Let's consider a specific case. Suppose you want to create a perpetuity with cash flows that grow by 2% per year, and you invest \$100 in a bank account that pays 5% interest. At the end of one year, you will have \$105 in the bank—your original \$100 plus 5% in interest. If you withdraw only \$3, you will have \$102 to reinvest—2% more than the amount you had initially. This amount will then grow to $102 \times 1.05 = \$107.10$ in the following year, and you can withdraw $\$3 \times 1.02 = \3.06 , which will leave you with principal of $\$107.10 - \$3.06 = \$104.04$. Note that $\$102 \times 1.02 = \104.04 . That is, both the amount you withdraw and the principal you reinvest grow by 2% each year. On a timeline, these cash flows look like this:



By following this strategy, you have created a growing perpetuity that starts at \$3 and grows 2% per year. This growing perpetuity must have a present value equal to the cost of \$100.

We can generalize this argument. In the case of an equal-payment perpetuity, we deposited an amount P in the bank and withdrew the interest each year. Because we always left the principal P in the bank, we could maintain this pattern forever. If we want to increase the amount we withdraw from the bank each year by g , then the principal in the bank will have to grow by the same factor g . So, instead of withdrawing all of the interest rP , we leave gP in the bank in addition to our original principal P , and only withdraw $C = (r - g)P$. Solving this last equation for P , the initial amount deposited in the bank account, gives the present value of a growing perpetuity with initial cash flow C :

Present Value of a Growing Perpetuity

$$PV(\text{growing perpetuity}) = \frac{C}{r - g} \quad (4.11)$$

To understand the formula for a growing perpetuity intuitively, start with the formula for a perpetuity. In the earlier case, you had to put enough money in the bank to ensure that the interest earned matched the cash flows of the regular perpetuity. In the case of a growing perpetuity, you need to put more than that amount in the bank because you have to finance the growth in the cash flows. How much more? If the bank pays interest at a rate of 5%, then all that is left to take out if you want to make sure the principal grows 2% per year is the difference: $5\% - 2\% = 3\%$. So instead of the present value of the perpetuity being the first cash flow divided by the interest rate, it is now the first cash flow divided by the *difference* between the interest rate and the growth rate.

EXAMPLE 4.10

Endowing a Growing Perpetuity

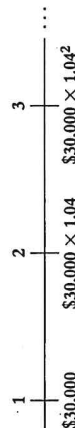
Problem

In Example 4.7, you planned to donate money to your alma mater to fund an annual \$30,000 MBA graduation party. Given an interest rate of 8% per year, the required donation was the present value of

$$PV = \$30,000/0.08 = \$375,000 \text{ today}$$

Before accepting the money, however, the MBA student association has asked that you increase the donation to account for the effect of inflation on the cost of the party in future years. Although \$30,000 is adequate for next year's party, the students estimate that the party's cost will rise by 4% per year thereafter. To satisfy their request, how much do you need to donate now?

Solution

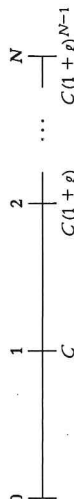


The cost of the party next year is \$30,000, and the cost then increases 4% per year forever. From the timeline, we recognize the form of a growing perpetuity. To finance the growing cost, you need to provide the present value today of

$$PV = \$30,000/(0.08 - 0.04) = \$750,000 \text{ today}$$

You need to double the size of your gift!

Growing Annuity. A growing annuity is a stream of N growing cash flows, paid at regular intervals. It is a growing perpetuity that eventually comes to an end. The following timeline shows a growing annuity with initial cash flow C , growing at rate g every period until period N :



The conventions used earlier still apply: (1) The first cash flow arrives at the end of the first period, and (2) the first cash flow does not grow. The last cash flow therefore reflects only $N - 1$ periods of growth.

The present value of an N -period growing annuity with initial cash flow C , growth rate g , and interest rate r is given by

Present Value of a Growing Annuity

$$PV = C \times \frac{1}{r - g} \left(1 - \left(\frac{1 + g}{1 + r} \right)^N \right) \quad (4.12)$$

Because the annuity has only a finite number of terms, Eq. 4.12 also works when $g > r$.⁷ The process of deriving this simple expression for the present value of a growing annuity is the same as for a regular annuity. Interested readers may consult the online appendix at www.berkdemarzo.com for details.

⁷ Equation 4.12 does not work for $g = r$. But in that case, growth and discounting cancel out, and the present value is equivalent to receiving all the cash flows at date 1: $PV = C \times N/(1 + r)$.

EXAMPLE 4.11

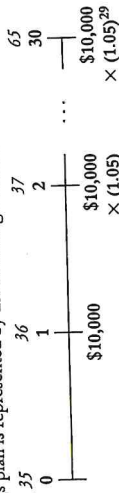
Retirement Savings with a Growing Annuity

Problem

In Example 4.9, Ellen considered saving \$10,000 per year for her retirement. Although \$10,000 is the most she can save in the first year, she expects her salary to increase each year so that she will be able to increase her savings by 5% per year. With this plan, if she earns 10% per year on her savings, how much will Ellen have saved at age 65?

Solution

Her new savings plan is represented by the following timeline:



This example involves a 30-year growing annuity, with a growth rate of 5%, and an initial cash flow of \$10,000. The present value of this growing annuity is given by

$$\begin{aligned} PV &= \$10,000 \times \frac{1}{0.10 - 0.05} \left(1 - \left(\frac{1.05}{1.10} \right)^{30} \right) \\ &= \$10,000 \times 15.0463 \\ &= \$150,463 \text{ today} \end{aligned}$$

Ellen's proposed savings plan is equivalent to having \$150,463 in the bank today. To determine the amount she will have at age 65, we need to move this amount forward 30 years:

$$\begin{aligned} FV &= \$150,463 \times 1.10^{30} \\ &= \$2.625 \text{ million in 30 years} \end{aligned}$$

Ellen will have saved \$2.625 million at age 65 using the new savings plan. This sum is almost \$1 million more than she had without the additional annual increases in savings.

The formula for the growing annuity encompasses all of the other formulas in this section. To see how to derive the other formulas from this one, first consider a growing perpetuity. It is a growing annuity with $N = \infty$. If $g < r$, then

$$\frac{1+g}{1+r} < 1, \text{ and so } \left(\frac{1+g}{1+r} \right)^N \rightarrow 0 \text{ as } N \rightarrow \infty.$$

The formula for a growing annuity when $N = \infty$ therefore becomes

$$PV = \frac{C}{r-g} \left(1 - \left(\frac{1+g}{1+r} \right)^N \right) = \frac{C}{r-g} (1-0) = \frac{C}{r-g},$$

which is the formula for a growing perpetuity. The formulas for a regular annuity and perpetuity also follow from the formula if we let the growth rate $g = 0$. So, if you remember the growing annuity formula, you've got them all!

CONCEPT CHECK

1. How do you calculate the present value of (a) a perpetuity, (b) an annuity, (c) growing perpetuity, and (d) growing annuity?
2. How are the formulas for the present value of a perpetuity, annuity, growing perpetuity, and growing annuity related?

4.6 Using an Annuity Spreadsheet or Calculator

Spreadsheet programs such as Excel, as well as common financial calculators, have a set of functions that perform the calculations that finance professionals do most often. In Excel, the functions are called **NPER**, **RATE**, **PV**, **PMT**, and **FV**. The functions are all based on the timeline of an annuity:



The interest rate used to discount these cash flows is denoted by **RATE**. Thus, there are a total of five variables: **NPER**, **RATE**, **PV**, **PMT**, and **FV**. Each function takes four of these variables as inputs and returns the value of the fifth one that ensures that the NPV of the cash flows is zero. That is, the functions all solve the problem

$$NPV = PV + PMT \times \frac{1}{RATE} \left(1 - \frac{1}{(1 + RATE)^{NPER}} \right) + \frac{FV}{(1 + RATE)^{NPER}} = 0 \quad (4.13)$$

In words, the present value of the annuity payments **PMT**, plus the present value of the final payment **FV**, plus the initial amount **PV**, has a net present value of zero. Let's tackle a few examples.

EXAMPLE 4.12

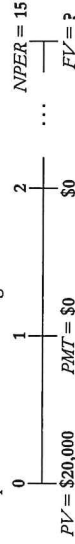
Computing the Future Value in Excel

Problem

Suppose you plan to invest \$20,000 in an account paying 8% interest. How much will you have in the account in 15 years?

Solution

We represent this problem with the following timeline:



To compute the solution, we enter the four variables we know (**NPER** = 15, **RATE** = 8%, **PV** = 20,000, **PMT** = 0) and solve for the one we want to determine (**FV**) using the Excel function **FV(RATE, NPER, PMT, PV)**. The spreadsheet here calculates a future value of \$63,443.

	NPER	RATE	PV	PMT	FV	Excel Formula
Given	15	8.00%	20,000	0		
Solve for FV					-63,443	=FV(0.08,15,0,20000)

Here we entered the **PV** as a positive number (the amount we are investing), and therefore the **FV** is shown as a negative number (the amount we can withdraw in the future). While the initial sign choice is arbitrary, when using an annuity calculator it is important to *change* the sign of the cash flow when money flows in the *opposite* direction.

To check the result, we can solve this problem directly:

$$FV = \$20,000 \times 1.08^{15} = \$63,443$$

The Excel spreadsheet in Example 4.12, which is available from www.berkdemarzo.com, is set up to allow you to compute any one of the five variables. We refer to this spreadsheet as the **annuity spreadsheet**. You simply enter the four input variables on the top line and leave the variable you want to compute blank. The spreadsheet computes the fifth variable and displays the answer on the bottom line. The spreadsheet also displays the Excel function that is used to get the answers. Let's work through a more complicated example that illustrates the convenience of the annuity spreadsheet.

EXAMPLE 4.13 Using the Annuity Spreadsheet

Problem

Suppose that you invest \$20,000 in an account paying 8% interest. You plan to withdraw \$2000 at the end of each year for 15 years. How much money will be left in the account after 15 years?

Solution

Again, we start with the timeline showing our initial deposit and subsequent withdrawals:

$$\begin{array}{ccccccc}
 0 & & 1 & & 2 & & \dots & & NPER = 15 \\
 PV = \$20,000 & & PMT = -\$2000 & & -\$2000 & & -\$2000 & & FV = ?
 \end{array}$$

Because we entered the *PV* as positive (the amount *invested*), we must enter the *PMT* as negative (the amount *withdrawn*). We solve for the final balance in the account, *FV*, using the annuity spreadsheet:

	NPER	RATE	PV	PMT	FV	Excel Formula
Given	15	8.00%	20,000	-2000		
Solve for FV					-9139	=FV(0.08,15,-2000,20000)

We will have \$9139 left in the bank to withdraw after 15 years.

We can also compute this solution directly. One approach is to think of the deposit and the withdrawals as being separate accounts. In the account with the \$20,000 deposit, our savings will grow to \$63,443 in 15 years, as we computed in Example 4.12. Using the formula for the future value of an annuity, if we borrow \$2000 per year for 15 years at 8%, at the end our debt will have grown to

$$\$2000 \times \frac{1}{0.08} (1.08^{15} - 1) = \$54,304$$

After paying off our debt, we will have \$63,443 - \$54,304 = \$9139 remaining after 15 years.

In addition to the annuity spreadsheet, an annuity calculator is available on MyLab (and is integrated into the eTextbook) which also shows the underlying calculations and the intermediate cash flows (as the box explains). You can also use a handheld financial calculator to do the same calculations. The calculators work in much the same way as the annuity spreadsheet. You enter any four of the five variables, and the calculator calculates the fifth variable.

1. What tools can you use to simplify the calculation of present values?
2. What is the process for using the annuity spreadsheet?

CONCEPT CHECK

Annuity Calculator

The annuity calculator tool is available in MyLab (and is integrated into the eTextbook) and works the same way as the annuity spreadsheet: Enter any four inputs and select the value you would like to solve for. In addition to the solution, you can display the amortization table which shows how the

balance in your account changes over time. Alternatively, you can choose to display the Excel formulas and calculations corresponding to Equation 4.13. Below we show the two modes (selected by the radio buttons at the bottom) for Example 4.13.

Annuity Calculator

Amortization table ☒ Calculations

Enter a decimal value or fraction (e.g., 5/12)	Number of payment periods	NPER	15
Enter an integer value or calculate the number of periods (e.g., 12 * 36)	Cash flow or annuity payment	PMT	-\$2,000.00
Enter a positive value for cash inflows and a negative value for cash outflows	Present value or current price	PV	\$20,000.00
Enter a positive value for cash inflows and a negative value for cash outflows	Future value or final value	FV	-\$9,139.15
Enter a positive value for cash inflows and a negative value for cash outflows	Instructions	Reset	Hide Data

Period	Starting Balance	Interest	Payment	Ending Balance
0	\$20,000.00	\$1,600.00	-\$2,000.00	\$19,600.00
1	\$19,600.00	\$1,568.00	-\$2,000.00	\$19,168.00
2	\$19,168.00	\$1,533.44	-\$2,000.00	\$18,701.44
3	\$18,701.44	\$1,496.12	-\$2,000.00	\$18,197.56
4	\$18,197.56	\$1,455.80	-\$2,000.00	\$17,653.36
5	\$17,653.36	\$1,412.27	-\$2,000.00	\$17,065.63
6	\$17,065.63	\$1,365.25	-\$2,000.00	\$16,430.88
7	\$16,430.88	\$1,314.47	-\$2,000.00	\$15,745.35
8	\$15,745.35	\$1,259.63	-\$2,000.00	\$15,004.98
9	\$15,004.98	\$1,200.40	-\$2,000.00	\$14,205.38
10	\$14,205.38	\$1,136.43	-\$2,000.00	\$13,341.81
11	\$13,341.81	\$1,067.34	-\$2,000.00	\$12,409.15
12	\$12,409.15	\$992.73	-\$2,000.00	\$11,401.88
13	\$11,401.88	\$912.15	-\$2,000.00	\$10,314.03
14	\$10,314.03	\$825.12	-\$2,000.00	\$9,139.15
15			-\$2,000.00	

Annuity Calculator

Amortization table ☐ Calculations

Annuity Equation:

$$0 = PV + \sum_{t=1}^N \frac{PMT}{(1+r)^t} + \frac{FV}{(1+r)^N}$$

$$0 = PV + PMT \times \frac{1}{r} \left(1 - \left(\frac{1}{1+r} \right)^N \right) + \frac{FV}{(1+r)^N}$$

Solution:

$$0 = \$20,000.00 + \frac{-\$2,000.00}{0.08} \left(1 - \frac{1}{1.08^{15}} \right) + \frac{FV}{1.08^{15}}$$

$$0 = \$20,000.00 + \frac{-\$2,000.00}{0.08} \left(1 - \frac{1}{1.08^{15}} \right) + \frac{FV}{1.08^{15}}$$

Excel:

$$FV(0.08, 15, -2000, 20000)$$

4.7 Non-Annual Cash Flows

Until now, we have only considered cash flow streams that occur at annual intervals. Do the same tools apply if the cash flows occur at another interval, say monthly? The answer is yes: Everything we have learned about annual cash flow streams applies to monthly cash flow streams so long as:

1. The interest rate is specified as a monthly rate.
2. The number of periods is expressed in months.

For example, suppose you have a credit card that charges 2% interest per month. If you have a \$1000 balance on the card today, and make no payments for six months, your future balance after six months will be

$$FV = C \times (1 + r)^n = \$1000 \times (1.02)^6 = \$1126.16$$

We apply the future value formula exactly as before, but with r equal to the *monthly* interest rate and n equal to the number of *months*.

The same logic applies to annuities, as in the following example.

EXAMPLE 4.14

Evaluating an Annuity with Monthly Cash Flows

Problem

You are about to purchase a new car and have two options to pay for it. You can pay \$20,000 in cash immediately, or you can get a loan that requires you to pay \$500 each month for the next 48 months (four years). If the monthly interest rate you earn on your cash is 0.5%, which option should you take?

Solution

Let's start by writing down the timeline of the loan payments:



The timeline shows that the loan is a 48-period annuity. Using the annuity formula the present value is

$$PV/(48\text{-period annuity of } \$500) = \$500 \times \frac{1}{0.005} \left(1 - \frac{1}{1.005^{48}} \right) = \$21,290$$

Alternatively, we may use the annuity spreadsheet to solve the problem:

	NPER	RATE	PV	PMT	FV	Excel Formula
Given	48	0.50%		-500	0	
Solve for PV			21,290			=PV(0.005, 48, -500, 0)

Thus, taking the loan is equivalent to paying \$21,290 today. So, by paying \$20,000 in cash rather than taking the loan, you save the equivalent of \$1290 today.

CONCEPT CHECK

1. Do the present and future value formulas depend upon the cash flows occurring at annual intervals?
2. When cash flows occur at a non-annual interval, what interest rate must you use? What number of periods must you use?

4.8 Solving for the Cash Payments

So far, we have calculated the present value or future value of a stream of cash flows. Sometimes, however, we know the present value or future value, but do not know the cash flows. The best example is a loan—you know how much you want to borrow (the present value) and you know the interest rate, but you do not know how much you need to repay each year. Suppose you are opening a business that requires an initial investment of \$100,000. Your bank manager has agreed to lend you this money. The terms of the loan state that you will make equal annual payments for the next 10 years and will pay an interest rate of 8% with the first payment due one year from today. What is your annual payment?

The timeline for the loan looks like this:



The bank will give you \$100,000 today in exchange for 10 equal payments over the next decade. You need to determine the size of the payment C that the bank will require. For the bank to be willing to lend you \$100,000, the loan cash flows must have a present value of \$100,000 when evaluated at the bank's interest rate of 8%. That is,

$$100,000 = PV(10\text{-year annuity of } C \text{ per year, evaluated at the loan rate})$$

Using the formula for the present value of an annuity,

$$100,000 = C \times \frac{1}{0.08} \left(1 - \frac{1}{1.08^{10}} \right) = C \times 6.71$$

Solving this equation for C gives

$$C = \frac{100,000}{6.71} = \$14,903$$

You will be required to make 10 annual payments of \$14,903 in exchange for \$100,000 today. We can also solve this problem with the annuity spreadsheet:

	NPER	RATE	PV	PMT	FV	Excel Formula
Given	10	8.00%	100,000		0	
Solve for PMT				-14,903		=PMT(0.08, 10, 100000, 0)

In general, when solving for a loan payment, think of the amount borrowed (the loan principal) as the present value of the payments when evaluated at the loan rate. If the payments of the loan are an annuity, we can solve for the payment of the loan by inverting the

annuity formula. Writing this procedure formally, we begin with the timeline for a loan with principal P , requiring N periodic payments of C and interest rate r :



Setting the present value of the payments equal to the principal,

$$P = PV(\text{annuity of } C \text{ for } N \text{ periods}) = C \times \frac{1}{r} \left(1 - \frac{1}{(1+r)^N} \right)$$

Solving this equation for C gives the general formula for the loan payment in terms of the outstanding principal (amount borrowed), P ; interest rate, r ; and number of payments, N :

Loan or Annuity Payment

$$C = \frac{P}{\frac{1}{r} \left(1 - \frac{1}{(1+r)^N} \right)} \quad (4.14)$$

Note that the cash flow for a perpetuity is simply $C = rP$. Rewriting (4.14) as $C = rP \left(\frac{1}{1 - 1/(1+r)^N} \right)$, we can see that the payment for an annuity always exceeds the payment of the equivalent value perpetuity, which makes sense because the annuity will eventually end.

EXAMPLE 4.15

Computing a Loan Payment

Problem

Your biotech firm plans to buy a new DNA sequencer for \$500,000. The seller requires that you pay 20% of the purchase price as a down payment, but is willing to finance the remainder by offering a 48-month loan with equal monthly payments and an interest rate of 0.5% per month. What is the monthly loan payment?

Solution

Given a down payment of $20\% \times \$500,000 = \$100,000$, your loan amount is \$400,000. We start with the timeline (from the seller's perspective), where each period represents one month:



Using Eq. 4.14, we can solve for the loan payment, C , as follows:

$$C = \frac{P}{\frac{1}{r} \left(1 - \frac{1}{(1+r)^N} \right)} = \frac{400,000}{\frac{1}{0.005} \left(1 - \frac{1}{(1.005)^{48}} \right)} = \$9394$$

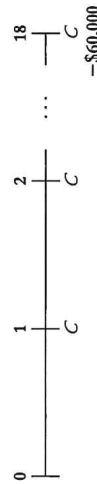
Using the annuity spreadsheet:

	NPER	RATE	PV	PMT	FV	Excel Formula
Given	48	0.50%	400,000		0	
Solve for PMT				-9,394		=PMT(0.005,48,400000,0)

Your firm will need to pay \$9394 each month to repay the loan.

We can use this same idea to solve for the cash flows when we know the future value rather than the present value. As an example, suppose you have just had a child. You decide to be prudent and start saving this year for her college education. You would like to have \$60,000 saved by the time your daughter is 18 years old. If you can earn 7% per year on your savings, how much do you need to save each year to meet your goal?

The timeline for this example is



That is, you plan to save some amount C per year, and then withdraw \$60,000 from the bank in 18 years. Therefore, we need to find the annuity payment that has a future value of \$60,000 in 18 years. Using the formula for the future value of an annuity from Eq. 4.10,

$$60,000 = FV(\text{annuity}) = C \times \frac{1}{0.07} (1.07^{18} - 1) = C \times 34$$

Therefore, $C = \frac{60,000}{34} = \1765 . So you need to save \$1765 per year. If you do, then at a 7% interest rate, your savings will grow to \$60,000 by the time your child is 18 years old.

Now let's solve this problem with the annuity spreadsheet:

	NPER	RATE	PV	PMT	FV	Excel Formula
Given	18	7.00%	0		-60,000	
Solve for PMT				1765		=PMT(0.07,18,0,-60000)

Once again, we find that we need to save \$1765 for 18 years to accumulate \$60,000.

CONCEPT CHECK

1. How can we solve for the required annuity payment for a loan?
2. How can we determine the required amount to save each year to reach a savings goal?

4.9 The Internal Rate of Return

In some situations, you know the present value and cash flows of an investment opportunity but you do not know the interest rate that equates them. This interest rate is called the **internal rate of return (IRR)**, defined as the interest rate that sets the net present value of the cash flows equal to zero.

For example, suppose that you have an investment opportunity that requires a \$1000 investment today and will have a \$2000 payoff in six years. On a timeline,



One way to analyze this investment is to ask the question: What interest rate, r , would you need so that the NPV of this investment is zero?

$$NPV = -1000 + \frac{2000}{(1+r)^6} = 0$$

Rearranging gives

$$1000 \times (1+r)^6 = 2000$$

That is, r is the interest rate you would need to earn on your \$1000 to have a future value of \$2000 in six years. We can solve for r as follows:

$$1+r = \left(\frac{2000}{1000} \right)^{1/6} = 1.1225$$

or $r = 12.25\%$. This rate is the IRR of this investment opportunity. Making this investment is like earning 12.25% per year on your money for six years.

When there are just two cash flows, as in the preceding example, it is easy to compute the IRR. Consider the general case in which you invest an amount P today, and receive FV in N years. Then the IRR satisfies the equation $P \times (1 + IRR)^N = FV$, which implies

$$IRR \text{ with two cash flows} = \left(\frac{FV}{P} \right)^{1/N} - 1 \quad (4.15)$$

Note in the formula that we take the total return of the investment over N years, FV/P , and convert it to an equivalent one-year return by raising it to the power $1/N$. Because we are just comparing two cash flows, the IRR calculation in Equation 4.15 is equivalent to computing the **compound annual growth rate (or CAGR)** of the cash flow.

Another case for which the IRR is easy to calculate is a perpetuity, as we demonstrate in the next example.

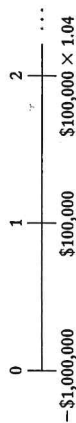
EXAMPLE 4.16 Computing the IRR for a Perpetuity

Problem

Jessica has just graduated with her MBA. Rather than take the job she was offered at a prestigious investment bank—Baker, Bellingham, and Botts—she has decided to go into business for herself. She believes that her business will require an initial investment of \$1 million. After that, it will generate a cash flow of \$100,000 at the end of one year, and this amount will grow by 4% per year thereafter. What is the IRR of this investment opportunity?

Solution

The timeline is



The timeline shows that the future cash flows are a growing perpetuity with a growth rate of 4%. Recall from Eq. 4.11 that the PV of a growing perpetuity is $C/(r-g)$. Thus, the NPV of this investment would equal zero if

$$1,000,000 = \frac{100,000}{r - 0.04}$$

We can solve this equation for r

$$r = \frac{100,000}{1,000,000} + 0.04 = 0.14$$

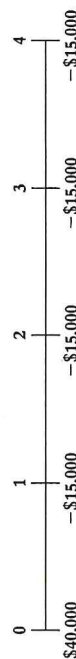
So, the IRR on this investment is 14%.

More generally, if we invest P and receive a perpetuity with initial cash flow C and growth rate g , we can use the growing perpetuity formula to determine

$$IRR \text{ of growing perpetuity} = (C/P) + g \quad (4.16)$$

Now let's consider a more sophisticated example. Suppose your firm needs to purchase a new forklift. The dealer gives you two options: (1) a price for the forklift if you pay cash and (2) the annual payments if you take out a loan from the dealer. To evaluate the loan that the dealer is offering you, you will want to compare the rate on the loan with the rate that your bank is willing to offer you. Given the loan payment that the dealer quotes, how do you compute the interest rate charged by the dealer?

In this case, we need to compute the IRR of the dealer's loan. Suppose the cash price of the forklift is \$40,000, and the dealer offers financing with no down payment and four annual payments of \$15,000. This loan has the following timeline:



From the timeline it is clear that the loan is a four-year annuity with a payment of \$15,000 per year and a present value of \$40,000. Setting the NPV of the cash flows equal to zero requires that the present value of the payments equals the purchase price:

$$40,000 = 15,000 \times \frac{1}{r} \left(1 - \frac{1}{(1+r)^4} \right)$$

The value of r that solves this equation, the IRR, is the interest rate charged on the loan. Unfortunately, in this case, there is no simple way to solve for the interest rate r .⁸ The only way to solve this equation is to guess values of r until you find the right one.

Start by guessing $r = 10\%$. In this case, the value of the annuity is

$$15,000 \times \frac{1}{0.10} \left(1 - \frac{1}{(1.10)^4} \right) = 47,548$$

The present value of the payments is too large. To lower it, we need to use a higher interest rate. We guess 20% this time:

$$15,000 \times \frac{1}{0.20} \left(1 - \frac{1}{(1.20)^4} \right) = 38,831$$

Now the present value of the payments is too low, so we must pick a rate between 10% and 20%. We continue to guess until we find the right rate. Let us try 18.45%:

$$15,000 \times \frac{1}{0.1845} \left(1 - \frac{1}{(1.1845)^4} \right) = 40,000$$

The interest rate charged by the dealer is 18.45%.

An easier solution than guessing the IRR and manually calculating values is to use a spreadsheet or calculator to automate the guessing process. When the cash flows are an annuity, as in this example, we can use the annuity spreadsheet in Excel to compute the IRR. Recall that the annuity spreadsheet solves Eq. 4.13. It ensures that the NPV of investing in the annuity is zero. When the unknown variable is the interest rate, it will solve for the interest rate that sets the NPV equal to zero—that is, the IRR. For this case,

	NPER	RATE	PV	PMT	FV	Excel Formula
Given	4		40,000	-15,000	0	
Solve for Rate		18.45%				=RATE(4, -15000, 40000, 0)

The annuity spreadsheet correctly computes an IRR of 18.45%.

EXAMPLE 4.17

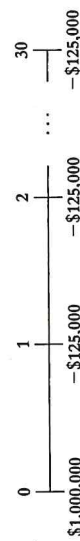
Computing the Internal Rate of Return for an Annuity

Problem

Baker, Bellingham, and Botts, was so impressed with Jessica that it has decided to fund her business. In return for providing the initial capital of \$1 million, Jessica has agreed to pay them \$125,000 at the end of each year for the next 30 years. What is the internal rate of return on Baker, Bellingham, and Botts's investment in Jessica's company, assuming she fulfills her commitment?

Solution

Here is the timeline of the loan:



⁸ With five or more periods and general cash flows, there is no general formula to solve for r ; trial and error (by hand or computer) is the only way to compute the IRR.

The timeline shows that the future cash flows are a 30-year annuity. Setting the NPV equal to zero requires

$$1,000,000 = 125,000 \times \frac{1}{r} \left(1 - \frac{1}{(1+r)^{30}} \right)$$

Using the annuity spreadsheet to solve for r ,

	NPER	RATE	PV	PMT	FV	Excel Formula
Given	30		1,000,000	-125,000	0	
Solve for Rate		12.09%				=RATE(30, -125000, 1000000, 0)

The IRR on this investment is 12.09%. In this case, we can interpret the IRR of 12.09% as the effective interest rate of the loan.

In this chapter, we developed the tools a financial manager needs to apply the NPV rule when cash flows occur at different points in time. As we have seen, the interest rate we use to discount or compound the cash flows is a critical input to any of our present or future value calculations. Throughout the chapter, we have taken the interest rate as given. What determines the interest rate that we should use when discounting cash flows? The Law of One Price implies that we must rely on market information to assess the value of cash flows across time. In Chapter 5, we learn the drivers of market interest rates as well as how they are quoted. Understanding interest rate quoting conventions will also allow us to extend the tools we developed in this chapter to situations where the cash flows are paid, and interest is compounded, more than once per year.

1. What is the internal rate of return?

2. In what two cases is the internal rate of return easy to calculate?

CONCEPT CHECK

USING EXCEL

Excel's IRR Function

Excel also has a built-in function, IRR, that will calculate the IRR of a stream of cash flows. Excel's IRR function has the format, IRR (values, guess), where "values" is the range containing the cash flows, and "guess" is an optional starting guess where Excel begins its search for an IRR. See the example below:

	A	B	C	D	E
1	Period	0	1	2	3
2	Cash Flow C_t	(1,000.0)	300.0	400.0	-500.0
3	IRR		8.9% =IRR(B2:E2)		

There are three things to note about the IRR function. First, the values given to the IRR function should include all of the cash flows of the project, including the one at date 0. In this sense, the IRR and NPV functions in Excel are inconsistent. Second, like the NPV function, the IRR ignores the period associated with any blank cells. Finally, as we will discuss in Chapter 7, in some settings the IRR function may fail to find a solution, or may give a different answer, depending on the initial guess.

See the eTextbook or MyLab Finance for an interactive IRR calculator.

Key Points and Equations

4.1 The Timeline

- Timelines are a critical first step in organizing the cash flows in a financial problem.

4.2 The Three Rules of Time Travel

- There are three rules of time travel:
 - Only cash flows that occur at the same point in time can be compared or combined.
 - To move a cash flow forward in time, you must compound it.
 - To move a cash flow backward in time, you must discount it.
- The future value in n years of a cash flow C today is

$$C \times (1 + r)^n \quad (4.1)$$
- The number of years it will take for an investment to double in value is approximately equal to 72 divided by the interest rate earned.
- The present value today of a cash flow C received in n years is

$$C + (1 + r)^n \quad (4.2)$$

4.3 Valuing a Stream of Cash Flows

- The present value of a cash flow stream is

$$PV = \sum_{n=0}^N \frac{C_n}{(1 + r)^n} \quad (4.4)$$
- The present value equals the amount you would need in the bank today to recreate the cash flow stream.
- The future value on date n of a cash flow stream with a present value of PV is

$$FV_n = PV \times (1 + r)^n \quad (4.5)$$

4.4 Calculating the Net Present Value

- The net present value (NPV) of an investment opportunity is PV (benefits – costs). The NPV is the net benefit of the investment in terms of an equivalent amount of cash today.

4.5 Perpetuities and Annuities

- A perpetuity is a constant cash flow C paid every period, forever. The present value of a perpetuity is

$$\frac{C}{r} \quad (4.7)$$

- An annuity is a constant cash flow C paid every period for N periods. The present value of an annuity is

$$C \times \frac{1}{r} \left(1 - \frac{1}{(1 + r)^N} \right) \quad (4.9)$$

The future value of an annuity at the end of the annuity is

$$C \times \frac{1}{r} \left((1 + r)^N - 1 \right) \quad (4.10)$$

- In a growing perpetuity or annuity, the cash flows grow at a constant rate g each period. The present value of a growing perpetuity is

$$\frac{C}{r - g} \quad (4.11)$$

The present value of a growing annuity is

$$C \times \frac{1}{r - g} \left(1 - \left(\frac{1 + g}{1 + r} \right)^N \right) \quad (4.12)$$

4.6 Using an Annuity Spreadsheet or Calculator

- Present and future values can be easily calculated using a spreadsheet program. Most programs have built-in formulas for evaluating annuities.

4.7 Non-Annual Cash Flows

- Monthly cash flow streams (or any other period length) can be evaluated in exactly the same way as annual cash flow streams so long as the interest rate and number of periods are expressed in monthly terms.

4.8 Solving for the Cash Payments

- The annuity and perpetuity formulas can be used to solve for the annuity payments when either the present value or the future value is known. The periodic payment on an N -period loan with principal P and interest rate r is

$$C = \frac{P}{\frac{1}{r} \left(1 - \frac{1}{(1 + r)^N} \right)} \quad (4.14)$$

4.9 The Internal Rate of Return

- The internal rate of return (IRR) of an investment opportunity is the interest rate that sets the NPV of the investment opportunity equal to zero.
- When there are only two cash flows, the IRR can be calculated as:

$$IRR \text{ with two cash flows} = \left(\frac{FV}{PV} \right)^{1/N} - 1 \quad (4.15)$$
- When the cash flows are a growing perpetuity with a starting cash flow of C with growth rate g , the IRR can be calculated as:

$$IRR \text{ of growing perpetuity} = (C/P) + g \quad (4.16)$$

Key Terms

- annuity *p. 151*
- annuity due *p. 153*
- annuity spreadsheet *p. 160*
- compound annual growth rate (CAGR) *p. 166*
- compound interest *p. 139*
- compounding *p. 139*
- consol *p. 149*
- discounting *p. 141*
- growing annuity *p. 157*
- growing perpetuity *p. 155*
- internal rate of return (IRR) *p. 166*
- net present value (NPV) *p. 147*
- perpetuity *p. 149*
- simple interest *p. 140*
- stream of cash flows *p. 137*
- time value of money *p. 139*
- timeline *p. 137*

Further Reading

The earliest known published work that introduces the ideas in this chapter was in 1202 by the famous Italian mathematician Fibonacci (or Leonardo of Pisa) in *Liber Abaci* (recently translated into English by Laurence Sigler, *Fibonacci's Liber Abaci: A Translation into Modern English of Leonardo Pisano's Book of Calculation*, Springer-Verlag, 2002). In this book, Fibonacci provides examples demonstrating the rules of time travel for cash flows.

Students who are interested in the early origins of finance and the historical development of the annuity formula will be interested in reading M. Rubinstein, *A History of the Theory of Investments: My Annotated Bibliography* (John Wiley and Sons, 2006) and W. Goetzmann and K. Rouwenhorst, eds., *The Origins of Value: The Financial Innovations that Created Modern Capital Markets* (Oxford University Press, 2005).

The material in this chapter should provide the foundation you need to understand the time value of money. For assistance using Excel, other spreadsheet programs, or financial calculators to compute present values, consult available help files and user manuals for additional information and examples. Students in the lucky position of having to decide how to receive lottery winnings may consult A. Atkins and E. Dyl, "The Lotto Jackpot: The Lump Sum versus the Annuity," *Financial Practice and Education* 5 (1995): 107–111.

Problems

The Timeline

1. You have just taken out a five-year loan from a bank to buy an engagement ring. The ring costs \$7000. You plan to put down \$1000 and borrow \$6000. You will need to make annual payments of \$1250 at the end of each year. Show the timeline of the loan from your perspective. How would the timeline differ if you created it from the bank's perspective?
2. You currently have a four-year-old mortgage outstanding on your house. You make monthly payments of \$1500. You have just made a payment. The mortgage has 26 years to go (i.e., it had an original term of 30 years). Show the timeline from your perspective. How would the timeline differ if you created it from the bank's perspective?

The Three Rules of Time Travel

3. Calculate the future value of \$2000 in
 - a. Five years at an interest rate of 5% per year.
 - b. Ten years at an interest rate of 5% per year.
 - c. Five years at an interest rate of 10% per year.
 - d. Why is the amount of interest earned in part (a) less than half the amount of interest earned in part (b)?
4. What is the present value of \$13,000 received
 - a. Fourteen years from today when the interest rate is 10% per year?
 - b. Twenty-eight years from today when the interest rate is 20% per year?
 - c. Seven years from today when the interest rate is 5% per year?
5. Your brother has offered to give you either \$75,000 today or \$150,000 in 9 years. If the interest rate is 8% per year, which option is preferable?
6. Consider the following alternatives:
 - i. \$150 received in one year
 - ii. \$220 received in five years
 - iii. \$330 received in ten years
 - a. Rank the alternatives from most valuable to least valuable if the interest rate is 11% per year.
 - b. What is your ranking if the interest rate is only 4% per year?
 - c. What is your ranking if the interest rate is 15% per year?

7. Suppose you invest \$1000 in an account paying 6% interest per year.
 - a. What is the balance in the account after 3 years? How much of this balance corresponds to "interest on interest"?
 - b. What is the balance in the account after 35 years? How much of this balance corresponds to interest on interest?
8. Your daughter is currently 11 years old. You anticipate that she will be going to college in seven years. You would like to have \$108,000 in a savings account to fund her education at that time. If the account promises to pay a fixed interest rate of 6% per year, how much money do you need to put into the account today to ensure that you will have \$108,000 in seven years?
9. You are thinking of retiring. Your retirement plan will pay you either \$250,000 immediately on retirement or \$350,000 five years after the date of your retirement. Which alternative should you choose if the interest rate is
 - a. 0% per year?
 - b. 8% per year?
 - c. 20% per year?
10. Your grandfather put some money in an account for you on the day you were born. You are now 18 years old and are allowed to withdraw the money for the first time. The account currently has \$3247 in it and pays a 6% interest rate.
 - a. How much money would be in the account if you left the money there until your 25th birthday?
 - b. What if you left the money until your 65th birthday?
 - c. How much money did your grandfather originally put in the account?

Valuing a Stream of Cash Flows

11. Suppose you receive \$100 at the end of each year for the next three years.
 - a. If the interest rate is 7%, what is the present value of these cash flows?
 - b. What is the future value in three years of the present value you computed in (a)?
 - c. Suppose you deposit the cash flows in a bank account that pays 7% interest per year. What is the balance in the account at the end of each of the next three years (after your deposit is made)? How does the final bank balance compare with your answer in (b)?
12. You have just received a windfall from an investment you made in a friend's business. He will be paying you \$46,819 at the end of this year, \$93,638 at the end of the following year, and \$140,457 at the end of the year after that (three years from today). The interest rate is 13.8% per year.
 - a. What is the present value of your windfall?
 - b. What is the future value of your windfall in three years (on the date of the last payment)?
13. You have a loan outstanding. It requires making six annual payments at the end of the next six years of \$9000 each. Your bank has offered to restructure the loan so that instead of making the six payments as originally agreed, you will make only one final payment at the end of the loan in six years. If the interest rate on the loan is 9.85%, what final payment will the bank require you to make so that it is indifferent between the two forms of payment?

Calculating the Net Present Value

14. You have been offered a unique investment opportunity. If you invest \$15,000 today, you will receive \$750 one year from now, \$2250 two years from now, and \$15,000 ten years from now.
 - a. What is the NPV of the opportunity if the interest rate is 12% per year? Should you take the opportunity?
 - b. What is the NPV of the opportunity if the interest rate is 8% per year? Should you take it now?
15. Marian Plunkett owns her own business and is considering an investment. If she undertakes the investment, it will pay \$8000 at the end of each of the next three years. The opportunity requires an